

LLaVA-MR: Large Language-and-Vision Assistant for Video Moment Retrieval

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Abstract. Multimodal Large Language Models (MLLMs) are widely used for visual perception, understanding, and reasoning. However, minutes-long video processing and precise moment retrieval remain challenging due to LLMs’ limited context size and coarse frame extraction. We propose the Large Language-and-Vision Assistant for Moment Retrieval (LLaVA-MR), which enables accurate moment retrieval and contextual grounding in videos using MLLMs. LLaVA-MR combines Dense Frame and Time Encoding (DFTE) for spatial-temporal feature extraction, Informative Frame Selection (IFS) for capturing brief visual and motion patterns, and Dynamic Token Compression (DTC) to manage LLM context limitations. Evaluations on benchmarks like Charades-STA, ActivityNet Cap and QVHighlights demonstrate that LLaVA-MR outperforms existing state-of-the-art methods, achieving an improvement of 1.82% in R1@0.5 and 1.29% in mAP@0.5 on the QVHighlights dataset. Our implementation will be open-sourced upon acceptance.

Keywords: Moment retrieval · temporal grounding.

1 Introduction

Large language models (LLMs) are widely recognized for their strong semantic understanding capabilities. Multimodal large language models (MLLMs) [7, 11], which extend LLMs to process multimodal data like video, have shown superior performance in tasks such as video description and visual question answering. They are also effective for specific visual tasks like object detection and visual grounding. However, **using MLLMs for *moment retrieval in minutes-long duration videos* is still a major challenge**, as it requires a deep understanding of visual changes and precise integration of temporal information.

Video Moment Retrieval (MR) aims to **locate specific temporal segments in untrimmed videos based on natural language queries**, requiring precise understanding of transient details. Prior methods [10, 19, 31, 34] rely on transformers to fuse image-text features and use task-specific heads to generate fixed moment candidates. However, their dependence on frame-level features limits comprehension, and specialized heads reduce robustness on new data. Approaches like [27]

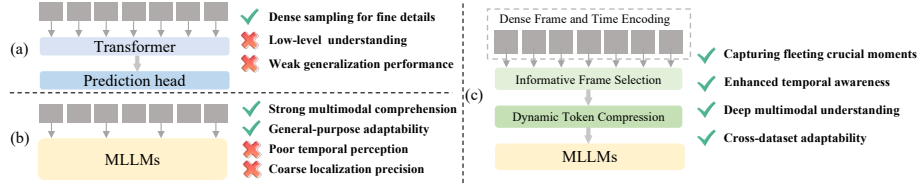


Fig. 1. Advantages of LLaVA-MR compared to prior works: (a)Transformer-based methods. (b) Previous MLLMs-based methods. (c) Our LLaVA-MR approach.

and [15] employ LLMs as assistants at the input stage but do not exploit their reasoning capabilities at output. Mr. BLIP [18] innovatively treats moment retrieval as a sequence-to-sequence generative task using MLLMs, enabling strong understanding and flexibility. However, MLLMs still face challenges in temporal perception, particularly in reliably associating sampled frames with their corresponding temporal positions [29, 8]. Besides, the limited context length of MLLMs necessitates sparse frame sampling, which can result in delayed detection of short-lived events or even complete omission of such events, thereby degrading localization precision.

We identify two key challenges limiting the effectiveness of MLLMs in moment retrieval: (1) improving their temporal awareness and perception to enable more precise alignment of video frames with time encoding, and (2) ensuring more accurate localization of brief moments within minutes-long videos. Due to limited context size, MLLMs often rely on sparse sampling, which reduces their ability to capture transient visual details in longer videos.

To address these challenges, we propose the **Large Language-and-Vision Assistant for Moment Retrieval (LLaVA-MR)**, which optimizes MLLMs for moment retrieval tasks by enhancing their ability to capture critical, brief moments within minutes-long video frame sequences, as illustrated in Fig. 1(c). To enhance temporal awareness and achieve higher-precision temporal perception, we apply Dense Frame and Time Encoding (DFTE) to extract fine-grained spatial and temporal features. To precisely locate brief moments in minutes-long videos, we developed an Informative Frame Selection (IFS) strategy to capture short-lived visual and motion patterns, and applied Dynamic Token Compression (DTC) to reduce sequence length while preserving essential information. This enables MLLMs to retrieve moments over extended video frame sequences effectively. Our approach achieves state-of-the-art results on widely used benchmarks, including Charades-STA [4], QVHighlights [10], and ActivityNet Captions [9].

In summary, our main contributions are the following:

- We propose LLaVA-MR, which applies MLLMs to the moment retrieval task, enhancing temporal awareness and effectively capturing critical, transient information in minutes-long videos.
- We introduce Dense Frame and Time Encoding (DFTE) to extract fine-grained spatial and temporal features, Informative Frame Selection (IFS) to

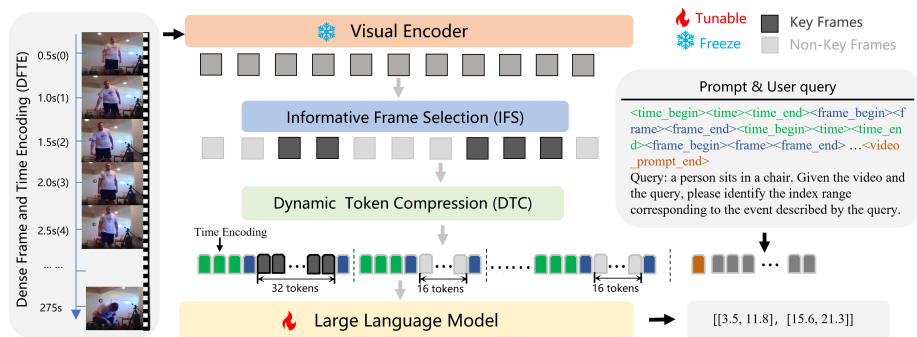


Fig. 2. LLaVA-MR model overview. We leverage a pretrained MLLM such as BLIP-2[12]. Our model primarily consists of Dense Frame and Time Encoding (DFTE), Informative Frame Selection (IFS), and Dynamic Token Compression (DTC).

capture critical dynamic moments, and Dynamic Token Compression (DTC) to compress redundant information.

- Our method achieves state-of-the-art performance across widely used public benchmarks, showing particularly strong advantages in moment retrieval.

2 Method

Our work enhances MLLMs for moment retrieval by improving temporal perception and capturing critical moments in minutes-long video sequences. § 2.1 provides an overview of the LLaVA-MR architecture. § 2.2 details the **Dense Frame and Time Encoding (DFTE)**, which extracts fine-grained visual features with time encoding for improved temporal awareness § 2.3 covers **Informative Frame Selection (IFS)** for capturing critical, brief moments within minutes-long video frame sequences. Finally, § 2.4 describes how the **Dynamic Token Compression (DTC)** reduces sequence length while preserving essential information. § 2.5 outlines the model training and inference process.

2.1 LLaVA-MR Architecture

Our model architecture is illustrated in Fig. 2. It starts with a frozen image encoder that extracts visual features from all sampled frames, generating a sequence of image embeddings defined as $\mathbf{F}^v = [\mathbf{f}_1^v, \dots, \mathbf{f}_N^v] \in \mathbb{R}^{N \times P \times D_0}$, where N represents the number of video frames, P is the number of patches, and D_0 is the visual feature dimension. We then apply Informative Frame Selection (IFS) to evaluate visual changes between frames, categorizing them as *key* or *non-key*. *Key* frames capture dynamic variations, often at event boundaries, while *non-key* frames show minimal change and carry more redundant information.

Next, **Q-Former**, a query-based transformer, refines each frame embedding by extracting relevant information. *Key frame* embeddings are denoted as $\mathbf{K}^q =$

$[\mathbf{f}_{i_1}^q, \mathbf{f}_{i_2}^q, \dots, \mathbf{f}_{i_k}^q] \in \mathbb{R}^{k \times Q \times D_1}$, and *non-key* frame embeddings are denoted as $\mathbf{N}^q = [\mathbf{f}_{j_1}^q, \mathbf{f}_{j_2}^q, \dots, \mathbf{f}_{j_{N-k}}^q] \in \mathbb{R}^{(N-k) \times Q \times D_1}$, where k is the number of *key* frames, Q represents the number of queries, and D_1 is the feature dimension after Q-Former. For *non-key* frames, we apply Dynamic Token Compression (DTC) within $\mathbf{N}^q$ to reduce sequence length while preserving essential information. Both $\mathbf{K}^q$ and the compressed $\mathbf{N}^q$ are projected into the language space, forming a unified embedding space, which is defined as $\mathbf{F}^L = [\mathbf{f}_1^L, \mathbf{f}_2^L, \dots, \mathbf{f}_N^L]$.

We then construct an interleaved multimodal input sequence consisting of time tokens t_i , frame embeddings $\mathbf{f}_i^L \in \mathbf{F}^L$, a query q , and a task prompt p . We position each time token directly before its corresponding frame embeddings, using special tokens T_B , T_E , F_B , and F_E to mark the start and end of the time and frame segments. Our interleaved multimodal sequence is structured as:

$$x = [T_B, t_1, T_E, F_B, \mathbf{f}_1^L, F_E, \dots, q, p] \quad (1)$$

where t_1 represents the first time token, and $\mathbf{f}_1^L$ represents the first frame embedding.

Finally, this sequence is fed into the LLM to predict the time segments relevant to query q . Following the approach of Mr. BLIP [18], the LLM is tasked with predicting the sequence of potentially multiple relevant moments, each defined by a start and end time in seconds. The prediction follows the format of a nested list of moments: $y = [[t_1^{\text{start}}, t_1^{\text{end}}], [t_2^{\text{start}}, t_2^{\text{end}}], \dots]$.

2.2 Dense Frame and Time Encoding

To enhance the temporal awareness of MLLMs, we propose selecting time encoding strategies based on the frame sampling rate, utilizing either relative frame indices or timestamps (in seconds). This choice helps avoid the token-splitting issues observed in previous work [18], which can cause confusion when large numbers or decimals are present. Based on this insight, we further observe that relative frame indices are more effective at high sampling rates, while timestamps in seconds offer clearer temporal references at low sampling rates.

N random frames are sampled from a uniform distribution over the interval $[0, T]$, where T represents the total video length. The distance between two consecutive random frames is given by $D = \text{frame}_i - \text{frame}_{i-1}$. Denoting the gaps between consecutive frames (including the start and end of the interval) as $G_1, \dots, G_{N+1}$, these gaps are uniformly distributed over all possible partitions of the total length. Consequently, each gap shares the same marginal distribution. For a Dirichlet distribution, the marginal distribution of any individual gap is $X = \frac{G_i}{T} \sim B(1, N)$, with the corresponding density function:

$$f_X(x) = N(1-x)^{N-1}, \quad 0 \leq x \leq 1 \quad (2)$$

By making the variable substitution $D = TX$, the density function of D is derived as:

$$f_D(d) = \frac{N}{T} \left(1 - \frac{d}{T}\right)^{N-1}, \quad 0 \leq d \leq T \quad (3)$$

Thus, the distance between two adjacent random frames follows the distribution $D \sim T \cdot B(1, N)$. For a Beta-distributed variable $X \sim B(1, N)$, the expectation is given by: $\mathbb{E}[X] = \frac{1}{1+N}$. Consequently, the expectation of D is:

$$\mathbb{E}[D] = \frac{T}{1+N} \quad (4)$$

When $\mathbb{E}[D] \leq 1$, the frames are considered densely sampled. For instance, if two random numbers fall within the interval $[1.5, 2.5]$, they are highly likely to be rounded to 2. In contrast, when $\mathbb{E}[D] > 1$, the frames are considered sparsely sampled. Finally, we define the *sampling rate* R_{frames} as:

$$R_{\text{frames}} = \frac{N+1}{T} \quad (5)$$

Two design scenarios are considered: (1) Dense Sampling Rate ($R_{\text{frames}} \geq 1$), we find that relative frame indices work best when frames are densely sampled, as each frame has a well-defined position in the sequence. In this setting, timestamps can be ambiguous due to rounding (e.g., 2.8s and 3.3s both round to 3s), causing information loss and potential model confusion. Relative frame indices provide a one-to-one mapping between frames and positions, preserving fine-grained temporal distinctions. (2) Low Sampling Rate ($R_{\text{frames}} < 1$), timestamps provide a more effective temporal reference. When frames are sparsely sampled, the temporal gap between consecutive frames is large. In this scenario, timestamps maintain a direct connection to absolute time than relative frame indices, which is crucial when temporal precision is required for moment retrieval tasks. § 3.3 will examine alternative temporal representations and analyze the impact of various time encoding strategies on model performance.

2.3 Informative Frame Selection

Dense sampling captures detailed visual information but adds redundancy, highlighting the need to distill critical visual and motion patterns. Brief transitions, like setting down a cup before picking up a phone, often hold more useful information, which is unevenly distributed across frames. To address this, we introduce the Informative Frame Selection (IFS) module, which captures brief yet crucial visual and motion variations.

In the IFS module, We categorize frames into two types: *key* frames and *non-key* frames. *Key* frames capture notable changes, often marking important event transitions and carrying valuable information. In contrast, *non-key* frames show minimal change, typically representing ongoing activities already covered by previous frames.

So how do we distinguish between *key* frames and *non-key* frames? For $\mathbf{F}^v$ obtained from the image encoder, we first compute the change between adjacent frames by subtracting each frame from the preceding one, denoted as $\Delta\mathbf{F}^v = [\Delta\mathbf{f}_1^v, \dots, \Delta\mathbf{f}_N^v] \in \mathbb{R}^{(N \times P \times D_0)}$, where each component represents the frame-to-frame variation across patches. For each $\Delta\mathbf{f}_i^v$ in $\Delta\mathbf{F}^v$, we then compute its L2

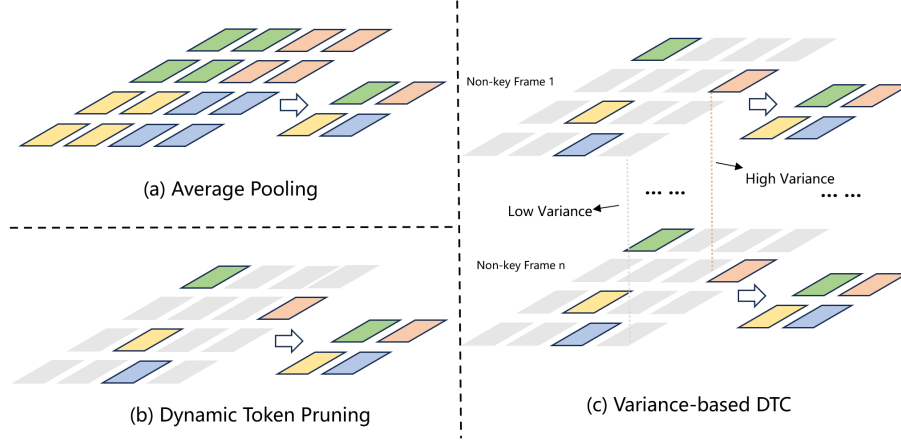


Fig. 3. Various token compression methods: (a) Average Pooling. (b) Dynamic Token Pruning, which reduces the number of tokens per image. (c) **Variance-Based Dynamic Token Compression (DTC)**, which ranks queries by variance across frames and retains tokens that focus on dynamic content.

norm to obtain $\mathbf{d} = [d_1, d_2, \dots, d_N]$. To further reduce the impact of video jitter, we apply a Gaussian filter to $\mathbf{d}$, resulting in the adjacent feature distance $\hat{\mathbf{d}}$, where each element $\hat{d}_i$ represents the change of the i -th frame relative to the previous one:

$$\hat{d}_i = \begin{cases} \max(\mathbf{d}), & \text{if } i = 0 \\ \mathcal{G}(\|\Delta \mathbf{f}_i^v\|_2), & \text{if } i > 0 \end{cases} \quad (6)$$

where, $\|\Delta \mathbf{f}_i^v\|_2$ denotes the L2 norm of the frame difference matrix $\Delta \mathbf{f}_i^v$, and $\mathcal{G}$ represents the Gaussian filter applied to smooth $\mathbf{d}$. During training, to maintain stability while preventing computational overload, we select the top- k frames with the highest change levels as *key* frames, while the remaining frames are classified as *non-key* frames. We can define this formally as follows:

$$\begin{cases} \mathbf{K}^v = [\mathbf{f}_m^v]_{m \in \text{top}_k(\hat{\mathbf{d}})} \\ \mathbf{N}^v = [\mathbf{f}_n^v]_{n \notin \text{top}_k(\hat{\mathbf{d}})} \end{cases} \quad (7)$$

where $\mathbf{K}^v$ represents the image embeddings of *key* frames, and $\mathbf{N}^v$ represents image embeddings of *non-key* frames. We will discuss the selection of the total frame number N and the setting of parameter k in § 3.3.

2.4 Dynamic Token Compression

As discussed in § 2.3, *non-key* frames typically exhibit fewer dynamic changes, often representing the repetitive or static continuation of an event. On the other hand, when frame sequences are too long, this not only reduces the performance of the LLM but also leads to significant resource consumption. It is important to reduce the sequence length while retaining valuable information.

Several token compression methods exist, such as simple pooling for down-sampling or dynamic token pruning [25], which adjust the token count per image. However, these methods overlook the relationships between frames, where tokens from different frames may be redundantly similar. Inspired by the design in [3] and [24], which encourage the model to capture more dispersed and diverse features from the input sequence, we propose a Variance-Based Dynamic Token Compression method (Variance-Based DTC).

After dividing the frames into *key* and *non-key* frames, we input $\mathbf{K}^v$ and $\mathbf{N}^v$ into the Q-Former to refine the embeddings. The refined embeddings for *key* and *non-key* frames are $\mathbf{K}^q \in \mathbb{R}^{k \times Q \times D_1}$ and $\mathbf{N}^q \in \mathbb{R}^{(N-k) \times Q \times D_1}$ (§ 2.1). Our variance-based DTC method aims to retain Q-Former queries that capture dynamic content in video frames. Since each Q-Former query focuses on different regions or aspects of the visual content, some queries, like those tracking static elements such as backgrounds, provide little guidance for predicting relevant results. To address this, we calculate the variance of each query across frames and retain only those with higher variance, which are more likely to capture dynamic aspects of the scene. The discarded tokens, having little variation between frames, can be compensated by the uncompressed tokens from the *key* frames. Section § 3.3 will discuss how different token compression methods affect our model performance.

2.5 Model Training and Inference

During training, we fine-tune our generative MLLM to tackle moment retrieval by reframing it as an open-ended, sequence-to-sequence task in natural language. The model is trained using a maximum likelihood objective. For each input x and target y , the loss is minimized as:

$$\mathcal{L}(x, y) = - \sum_{i=1}^{L-1} \log p_{\theta}(y_{i+1} | x, y_{1:i}), \quad (8)$$

where L is the length of the target sequence. We employ the LoRA [5] parameter-efficient fine-tuning approach to effectively adapt with minimal additional parameters by injecting low-rank matrices into the model layers.

For inference, we sample frames and use the text decoder to generate the moment sequence, guided by the model’s likelihood with beam search. The final sequence is output as a Python-style list of predicted moments, with minor formatting adjustments made as needed. Notably, when using relative frame indices as time encoding, we record the mapping between these indices and actual timestamps, and then convert the output back to specific timestamps.

3 Experiment

3.1 Experiment Setting

Benchmarks. We validate our LLaVA-MR on three widely used video moment retrieval benchmarks: Charades-STA [4], ActivityNet Captions [9], and QVHigh-

lights [10]. Charades-STA contains short daily activity videos with sentence-moment annotations; ActivityNet Captions is a large-scale dataset of long videos with dense descriptions; and QVHighlights targets highlight localization in user-generated videos. We follow the train/val/test splits and official evaluation protocols for all datasets.

Metrics. Common metrics for moment retrieval include Recall@K, mean Average Precision (mAP), and mean Intersection over Union (mIoU), evaluated at various IoU thresholds. Recall@K denotes the percentage of top-K predictions exceeding a given IoU (e.g., R1@0.5). mIoU measures average overlap, and mAP assesses precision at specific IoUs like 0.5 and 0.75. In short, R1@0.5 and R1@0.7 evaluate recall of top-1 predictions, mIoU reflects average intersection, and mAP captures precision across IoU levels.

Implementation details. Our model follows the Blip-2 [13] setup, using ViT-L/14 from CLIP [22] as the vision encoder and FlanT5 [2] as the LLM. Considering the relatively small scale of the datasets, we employ LoRA-based parameter-efficient finetuning to train only 0.63% of the total parameters in the model. Due to the output instability of LLMs, we mitigate minor formatting inconsistencies by applying post-processing heuristics to improve prediction accuracy. Our optimizer is AdamW [17], with an initial learning rate (LR) of 1e-8, linearly warmed up to 3e-4 over 10% of iterations, followed by cosine LR decay. Frames are sampled randomly during training.

For Charades-STA, we sample 60 frames per video and train for up to 50 epochs with a batch size of 16 across 4 GPUs, amounting to approximately 70 GPU hours. For QVHighlights Captions, we sample 80 frames per video, train for up to 50 epochs with a batch size of 32 on 8 GPUs, and set gradient accumulation to 4, resulting in about 33 GPU hours. For ActivityNet, we sample 80 frames per video and train for up to 50 epochs with a batch size of 32 across 8 GPUs, totaling around 70 GPU hours. All GPUs used are A100-80GB.

3.2 Results

Comparison to the State-of-the-Art. To evaluate our approach for moment retrieval, we compare LLaVA-MR with state-of-the-art methods on three widely used datasets: *Charades-STA*, *QVHighlights* and *Activity Captions*. As shown in Table 1, LLaVA-MR outperforms all other methods across nearly all key metrics on all three datasets. This demonstrates the superiority and robustness of our approach. Notably, on the QVHighlights dataset, which features longer videos requiring deeper understanding compared to Charades-STA and Activity Captions, our LLaVA-MR shows a more significant improvement. On the test set, it outperforms other methods with gains of **1.82%** in R1@0.5 and **1.29%** in mAP@0.5. These results highlight our model’s enhanced capability to handle moment retrieval tasks in longer, more complex videos.

Inference Time. We compared the inference times of LLaVA-MR and Mr. BLIP [18] with the same number of sampled frames. After averaging 10 tests on an A100-80G GPU, LLaVA-MR took 1.31s, while Mr. BLIP took 1.57s. The ad-



Fig. 4. Qualitative results on Charades-STA [4] and QVHighlights [10]: Predicted and ground truth segments are shown for comparison.

dition of modules such as IFS was offset by the compression mechanism, resulting in faster inference.

Qualitative Results. Fig. 4 visualizes qualitative results, with ground truth segments for query events alongside highlighted predicted intervals. In examples 1 and 2, LLaVA-MR accurately identifies the segments matching the natural language queries, with predicted start and end times closely aligning with ground truth. In example 3, the model not only outputs the correct segment but also detects an additional relevant segment that was not annotated.

3.3 Ablation Studies

To assess the impact of each component in our approach, we conducted a detailed ablation study using the Charades-STA dataset. The results are presented in Table 2. The baseline method (a) involves sparse sampling from the video and feeding it into the MLLM, which confirms the viability of using MLLM for the moment retrieval task. In experiment (b), we increased the number of sampled frames based on the baseline to capture more visual information, resulting in an improvement of **+0.89** in R1@0.5. Experiment (c) builds on (b) by optimizing temporal representation, thereby enhancing the model’s understanding of temporal sequences, which led to a further improvement of **+2.15** in R1@0.5. When all frames are compressed without IFS, (d) shows a performance decline due to the loss of valuable information. Finally, in experiment (e), we integrated the IFS and DTC modules to capture crucial moments and compress redundant information, achieving a significant improvement of **+2.52** in R1@0.5.

Design of Time Encoding. Designing an effective time encoding method is essential for enabling MLLMs to associate each frame with its corresponding time. Common approaches include using frame indices or timestamps to indicate each frame position. Frame indices can be either relative, representing the order within the sampled sequence (*e.g.*, 1, 2, 3, ..., 60), or absolute, marking the frame

Table 1. Comparison with State-of-the-Art on Charades-STA, QVHighlights, and ActivityNet Captions: **CLIP***: UnLoc [31] is pretrained on action classification datasets (Kinetics 400/700). **SF**: Slow-Fast backbone.

Method	Backbone	mIoU	R1@0.5	R1@0.7	mAP@0.5	mAP@0.75
QVHighlights Validation set						
Moment-DETR[10]	SF+CLIP	–	59.68	40.84	–	–
EaTR [6]	I3D	–	61.36	45.79	61.86	41.91
QD-DETR [20]	SF+CLIP	–	62.68	46.66	62.23	41.82
UVCOM [30]	SF+CLIP	–	65.10	51.81	–	–
UnLoc-L [31]	CLIP*	–	66.10	46.70	–	–
CG-DETR [19]	SF+CLIP	–	67.40	52.10	65.60	45.70
R^2 -Tunning [16]	CLIP	–	68.71	52.06	–	–
VideoLights-B-pt [21]	SF+CLIP+BLIP	–	72.06	57.94	70.38	51.12
FlashVTG [1]	SF+CLIP	–	73.10	57.29	72.75	54.33
Mr. BLIP [18]	BLIP-2	–	76.13	63.35	69.39	55.78
LLaVA-MR	CLIP+LLM	71.85	78.13	64.13	69.64	56.32
QVHighlights Test set						
SeViLa [32]	BLIP-2	–	54.50	36.50	–	–
UniVTG [14]	SF+CLIP	–	58.86	40.86	57.60	35.59
QD-DETR [20]	SF+CLIP	–	62.40	44.98	62.52	39.88
UVCOM [30]	SF+CLIP	–	63.55	47.47	63.37	42.67
LMR [15]	SF+CLIP	–	64.40	47.21	64.65	43.16
CG-DETR [19]	SF+CLIP	–	65.43	48.38	64.51	42.77
R^2 -Tunning [16]	CLIP	–	68.03	49.35	69.04	47.56
VideoLights-B-pt [21]	SF+CLIP+BLIP	–	70.36	55.25	69.53	49.17
FlashVTG [1]	SF+CLIP	–	70.69	53.96	72.33	53.85
Mr. BLIP [18]	BLIP-2	–	74.77	60.51	68.12	53.38
LLaVA-MR	CLIP+LLM	–	76.59	61.48	69.41	54.40
Charades-STA Test set						
Moment-DETR [10]	–	–	53.63	31.37	–	–
QD-DETR [20]	SF+CLIP	–	57.31	32.55	–	–
LMR [15]	SF+CLIP	–	55.91	35.19	–	–
UniVTG [14]	SF+CLIP	50.10	58.01	35.65	–	–
CG-DETR [19]	SF+CLIP	50.13	58.44	36.34	–	–
UVCOM [30]	SF+CLIP	–	59.25	36.64	–	–
R^2 -Tunning [16]	CLIP	–	59.78	37.02	–	–
UnLoc-L [31]	CLIP*	–	60.80	38.40	–	–
VideoLights-B-pt [21]	SF+CLIP+BLIP	52.94	61.96	41.05	–	–
UniMD+Sync [34]	–	–	63.98	44.46	–	–
InternVideo2-1B [28]	–	–	68.36	45.03	–	–
EaTR [6]	I3D	–	68.47	44.92	–	–
Mr. BLIP [18]	BLIP-2	58.63	69.31	49.29	–	–
InternVideo2-6B [28]	–	–	70.03	48.95	–	–
FlashVTG [1]	SF+CLIP	–	70.32	49.87	–	–
LLaVA-MR	CLIP+LLM	59.78	70.65	49.58	69.96	39.66
ActivityNet Captions						
DRN [33]	–	–	45.45	24.36	–	–
VLG-Net [23]	–	–	46.32	29.82	–	–
UnLoc-B [31]	CLIP*	–	48.00	29.70	–	–
UnLoc-L [31]	CLIP*	–	48.30	30.20	–	–
GVL [26]	–	–	49.18	29.69	–	–
Mr. BLIP [18]	BLIP-2	–	53.92	35.55	–	–
LLaVA-MR	CLIP+LLM	–	55.16	35.68	–	–

Table 2. Comprehensive Ablation Study. DF: Dense Frame, TimeEnc: Time Encoding.

	DF	TimeEnc	DTC	IFS&DTC	R1@0.5	R1@0.7	mIoU
(a)					65.48	42.71	55.17
(b)	✓				66.37	44.65	57.50
(c)	✓	✓			68.52	47.29	58.16
(d)	✓	✓	✓		68.13	47.05	58.11
(e)	✓	✓	✓	✓	70.65	49.58	59.78

Table 3. Ablation study of time encoding methods.

Time Encoding	R1@0.5	R1@0.7	mIoU
Charades-STA			
Relative indices	67.83	46.68	57.78
Timestamps	66.37	44.65	57.50
QVHighlights			
Relative indices	74.77	61.74	68.45
Timestamps	76.06	62.65	70.01

position in the full video (*e.g.*, 1, 15, 30, ..., 900). Since larger numbers are often split into multiple tokens during tokenization, leading to potential numerical confusion and lower performance for LLMs [18], we focus on relative indices over absolute ones. Similarly, directly inputting decimal timestamps can cause tokenization issues; thus, rounding to the nearest integer is a common approach, which we apply as well when using timestamps.

In Table 3, we validate the hypothesis that the choice of time representation depends on the sampling rate $\mathcal{R}_{frames}$ (see § 2.2). For Charades-STA, with average video length 30 seconds and 60 sampled frames ($\mathcal{R}_{frames} \approx \frac{60+1}{30} > 1$), using relative frame indices improves R1@0.5 by +**1.46** compared to timestamps. This is because timestamps rounded to integers can become identical (*e.g.*, 0.7s and 1.1s both round to 1s), confusing the model, whereas relative indices preserve clear sequential distinctions. Conversely, for QVHighlights, with 150 seconds average length and 80 sampled frames ($\mathcal{R}_{frames} \approx \frac{80+1}{150} < 1$), timestamps outperform relative indices by +**1.29** in R1@0.5. We attribute this to larger temporal gaps between frames, where relative indices lack precision (*e.g.*, index 1 could correspond to varying actual times), while timestamps provide more accurate temporal localization.

Additionally, to enhance the model’s ability to distinguish between tokens representing temporal and frame-based information, we introduced special tokens and evaluated their impact on Charades-STA, with results presented in Table 4, demonstrating their effectiveness in improving model performance.

Design of Informative Frame Selection. The Informative Frame Selection (IFS) module introduced in § 2.3 captures brief but critical visual and motion variations. This module **classifies frames into *key* and *non-key* categories by calculating the adjacent feature distance $\hat{\mathbf{d}}$** . We selected a sample video and plotted $\hat{\mathbf{d}}$, as shown in Fig. 5. It can be seen that $\hat{\mathbf{d}}$ peaks when the person picks up or puts down the guitar, while remaining low during the guitar playing. This supports the hypothesis that frames with more significant dynamic changes not only contain more valuable visual information but are also more likely to lie at event boundaries, while frames within event interiors remain relatively static.

Besides, we investigated two strategies for key-frame selection on the Charades-STA dataset: **fixed top- k (our current approach)** and a **distance-threshold method based on adjacent feature distance $\hat{\mathbf{d}}$** . In the threshold-based approach, any frame exceeding a set distance is marked as key, and we varied the threshold from 0.05 to 0.95 in steps of 0.05, with the best performance at 0.7. For top- k , the optimal

Table 4. Ablation study of prompt design on Charades-STA

Input Format	R1@0.5	R1@0.7	mIoU
No special tokens	67.83	46.68	57.78
B/E for time	67.98	46.80	57.93
B/E for frames	68.25	47.07	58.02
B/E for both	68.52	47.29	58.16

Table 5. Ablation study of different token compression methods

Compression Method	R1@0.5	R1@0.7	mIoU
No Compression	68.52	47.29	58.16
Average Pooling	69.43	48.17	58.77
Variance-Based DTC	70.65	49.58	59.78

Table 6. Comparison of different selection strategies

Strategies in IFS	R1@0.5	R1@0.7	mIoU
top- k	70.65	49.58	59.78
distance-threshold	69.98	48.72	58.67

Table 7. Impact of IFS&DTC modules on VideoLLaMA

Model	R1@0.5	R1@0.7	mIoU
VideoLLaMA (original)	48.93	27.64	31.27
VideoLLaMA + IFS&DTC	53.26	30.12	32.94

result was achieved with $N = 60, k = 32$. As summarized in Table 6, fixed top- k consistently outperforms threshold-based selection. We attribute this to video diversity: threshold-based selection can yield too few or too many key frames, while top- k ensures a consistent number, better balancing informativeness and efficiency.

Moreover, we analyze the impact of **total frames N** and top- k selection on Charades-STA (Fig. 6). Too small k causes excessive compression and information loss, while too large k adds redundancy, both harming performance. Similarly, increasing N beyond an optimal value introduces additional redundancy without improvement. We find **$N = 60, k = 32$ optimal for Charades-STA** and $N = 80, k = 32$ for QVHighlights. Further experiments and hyperparameter details are provided in the supplementary material.

Design of Dynamic Token Compression. After dividing the frames into *key* and *non-key* frames, we employ the Dynamic Token Compression (DTC) module from § 2.4 to compress the features of non-key frames, thereby reducing the sequence length and eliminating redundancy. This compression strategy is designed to retain essential information while filtering out redundant data.

Table 5 shows the impact of different compression methods on the model performance. Without using any compression strategy, the model achieves a score of **68.52** on R1@0.5. We first applied average pooling to N^q , reducing the token corresponding to each frame by half. This helped reduce redundancy and shorten the sequence length, achieving a score of **69.43** on R1@0.5. However, we infer that simply applying average pooling does not guarantee that more valuable information in N^q is preserved. This led us to the design of Variance-Based DTC, with an aim to retain Q-Former queries that capture dynamic content in the video frames. Variance-Based DTC outperformed average pooling, improving R1@0.5 to **70.65**. We attribute this improvement to the fact that queries with higher variance are more likely to capture dynamic aspects of the scene, while lower-variance queries correspond to relative stable features across

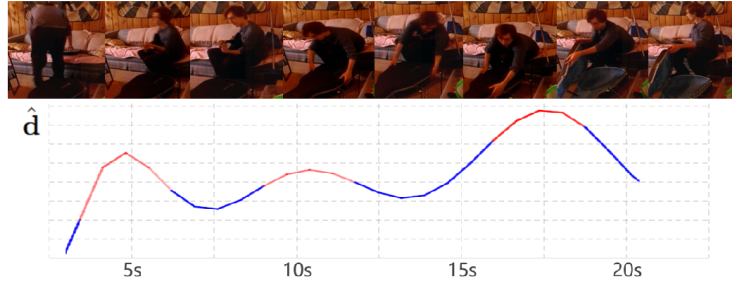


Fig. 5. Temporal correspondence between video frames and adjacent frame feature distance $\hat{d}$.

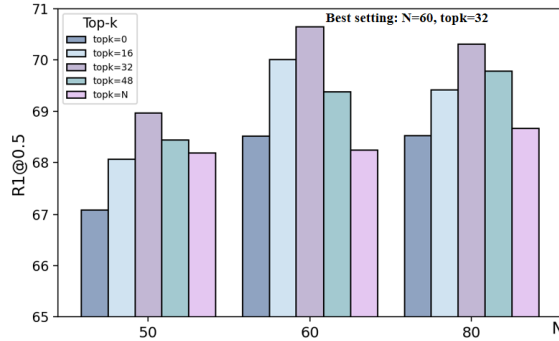


Fig. 6. Hyperparameter optimization results on Charades-STA to determine the optimal values for N and k during training.

frames and can be more effectively complemented by features from other frames after compression.

Generalization to Other MLLMs. To assess the applicability of our proposed modules to different MLLMs, we integrated IFS and DTC into VideoLLaMA[35] and evaluated performance on Charades-STA. As reported in Table 7, the inclusion of these modules consistently improves moment retrieval metrics, indicating that IFS and DTC can provide effective enhancements beyond the base model.

4 Conclusion

We introduce LLaVA-MR, a novel approach that optimizes MLLMs for moment retrieval in minutes-long videos by enhancing their capability to capture brief but critical moments within extensive sequences. Through comprehensive evaluations, LLaVA-MR demonstrates significant performance improvements, outperforming ten leading methods, especially on complex, minutes-long video benchmarks like QVHighlights.

Limitations. The current approach mainly uses video frames and text queries, neglecting other modalities like audio, which limits the model’s ability to fully capture the rich contextual information. Additionally, the prompt format is restricted to text, narrowing the range of possible queries and real-world applications.

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